

Quantification of the neurotoxic beta-carboline harmane in barbecued/grilled meat samples and correlation with level of doneness.



Harmane, one of the heterocyclic amines (HCAs), is a potent neurotoxin linked to human diseases. Dietary exposure, especially in cooked meats, is the major source of exogenous exposure for humans. However, knowledge of harmane concentrations in cooked meat samples is limited. Our goals were to (1) quantify the concentration of harmane in different types of cooked meat samples, (2) compare its concentration to that of other more well-understood HCAs, and (3) examine the relationship between harmane concentration and level of doneness. Thirty barbecued/grilled meat samples (8 beef steak, 12 hamburger, 10 chicken) were analyzed for harmane and four other HCAs (2-amino-1-methyl-6-phenylimidazo [4,5-b]pyridine [PhIP], amino-3,8-dimethylimidazo[4,5-f]quinoxaline [MeIQx], 2-amino-3,4,8-trimethylimidazo[4,5-f]quinoxaline [DiMeIQx], and 2-amino-1,6-dimethylfuro[3,2-e]imidazo[4,5-b]pyridine [IFP]). Mean (+/- SD) harmane concentration was 5.63 (+/- 6.63) ng/g; harmane concentration was highest in chicken (8.48 +/- 9.86 ng/g) and lowest in beef steak (3.80 +/- 3.6 ng/g). Harmane concentration was higher than that of the other HCAs and significantly correlated with PhIP concentration. Harmane concentration was associated with meat doneness in samples of cooked beef steak and hamburger, although the correlation between meat doneness and concentration was greater for PhIP than for harmane. Evidence indicates that harmane was detectable in nanograms per gram quantities in cooked meat (especially chicken) and, moreover, was more abundant than other HCAs. There was some correlation between meat doneness and harmane concentration, although this correlation was less robust than that observed for PhIP. Data such as these may be used to improve estimation of human dietary exposure to this neurotoxin.